

1. Error and inverse stability

1. Pointwise bias-variance decomposition

Fix x_0 and let the random training sample S produce $\hat{f}_{S(x_0)}$. A fresh response is

$$Y = f^*(x_0) + \varepsilon, \quad f^*(x_0) = \mathbb{E}[Y|x_0].$$

Assume squared loss, $\mathbb{E}[\varepsilon|x_0] = 0$, $\text{Var}(\varepsilon|x_0) = \sigma^2(x_0)$, and fresh ε independent of S . With $\mu(x_0) = \mathbb{E}_S[\hat{f}_{S(x_0)}]$,

$$\mathbb{E}_{S,\varepsilon} \left[(Y - \hat{f}_{S(x_0)})^2 \right] = (\mu - f^*)^2 + \text{Var}_{S(\hat{f}_{S(x_0)})} + \sigma^2(x_0).$$

The three terms are squared systematic displacement, refit spread, and fresh noise. The identity is **pointwise**: x_0 is fixed.

2. Why the cross terms vanish

At the fixed point, write $Y - \hat{f} = (f^* - \mu) + (\mu - \hat{f}) + \varepsilon$.

The training-sample cross term is zero because $\mathbb{E}_S[\hat{f}] = \mu$. For a fresh-noise cross term, condition first on the sample:

$$\mathbb{E}[(\mu - \hat{f})\varepsilon] = \mathbb{E}_S[(\mu - \hat{f})\mathbb{E}[\varepsilon|S, x_0]] = 0.$$

Freshness plus zero conditional mean supplies $\mathbb{E}[\varepsilon|S, x_0] = 0$. $\mathbb{E}[\varepsilon] = 0$ alone does **not** rule out correlation with the fitted predictor; if freshness fails, extra terms may remain.

3. Hadamard audit: three separate tests

An inverse problem is well posed on its declared admissible spaces only if:

1. a solution exists for every admissible observation;
2. the solution is unique in the modeled solution class;
3. the solution depends continuously on the observation.

Failure of any one makes the problem ill posed. For $\text{diag}(1, 0)$, uniqueness or existence can fail. For $\text{diag}(1, 0.01)$, the inverse exists and is continuous, but has norm 100: severe sensitivity, not discontinuity. **Zero** and **small positive** singular values are qualitatively different.

4. Singular-value conditioning

For singular pairs $Av_j = \sigma_j u_j$, $\sigma_j \geq 0$, every active direction $\sigma_j > 0$ is inverted by

$$\langle f, v_j \rangle = \frac{\langle y, u_j \rangle}{\sigma_j}.$$

An observation perturbation in direction u_j is amplified by $\frac{1}{\sigma_j}$. For invertible A ,

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}.$$

κ_2 is a worst-case **relative** sensitivity measure; $\frac{1}{\sigma_{\min}}$ is an **absolute** inverse-amplification bound. A condition number is not the statistical variance from the bias-variance decomposition, though small singular directions provide a mechanism for refit instability.

2. Tikhonov regularization

1. Objective and finite-dimensional solution

For $A : \mathcal{H} \rightarrow \mathcal{K}$, observation y , and $\lambda > 0$, zero-prior Tikhonov solves

$$f_\lambda \in \arg \min_f \{ \|Af - y\|^2 + \lambda \|f\|^2 \}.$$

The first term is data misfit; the second charges for solution size.

Increasing λ changes their relative weight; it does not remove noise from y .

For $A \in \mathbb{R}^{n \times d}$ with Euclidean norms,

$$(A^T A + \lambda I) f_\lambda = A^T y.$$

For nonzero z , $z^T (A^T A + \lambda I) z = \|Az\|^2 + \lambda \|z\|^2 > 0$. Thus the finite-dimensional linear quadratic has exactly one minimizer even when A has a null space. This proof does not establish a general nonlinear or infinite-dimensional existence theorem.

2. Directionwise filter and stability

Using the positive compact-SVD pairs $Av_j = \sigma_j u_j$,

$$f_\lambda = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \lambda} \langle y, u_j \rangle v_j, \quad P_{\ker(A)} f_\lambda = 0.$$

- unregularized data-to-solution gain: $\frac{1}{\sigma_j}$;
- regularized gain: $\frac{\sigma_j}{\sigma_j^2 + \lambda}$;
- fraction of the unregularized coefficient retained: $\frac{\sigma_j^2}{\sigma_j^2 + \lambda}$.

The zero-prior penalty sets the separate null-space component to zero; it does not recover missing information. For fixed $\lambda > 0$,

$$\|f_{\lambda(y+\Delta y)} - f_{\lambda(y)}\|_2 \leq \frac{1}{2\sqrt{\lambda}} \|\Delta y\|_2,$$

because $\max_{\sigma \geq 0} \frac{\sigma}{\sigma^2 + \lambda} = \frac{1}{2\sqrt{\lambda}}$.

3. Zero-penalty boundary

In finite-dimensional linear least squares with the Euclidean zero-prior penalty,

$$f_\lambda \rightarrow A^\dagger y \quad \text{as } \lambda \rightarrow 0^+.$$

Positive singular coordinates approach the inverse coefficients, while every null-space coordinate remains zero. The limit is therefore the Moore-Penrose minimum-norm least-squares solution: the minimum-norm exact solution if the system is consistent, or the minimum-norm member among least-squares minimizers otherwise. Under full column rank this reduces to the unique OLS solution.

For general inverse problems, sending λ to zero is not enough. A representative noise-compatible schedule also requires $\frac{\delta^2}{\lambda} \rightarrow 0$, together with the relevant analytic assumptions. Fixed positive λ is not an exact-recovery limit.

3. Ridge specialization

1. Convention, objective, estimator

Let $X \in \mathbb{R}^{n \times d}$ be fixed, with centered columns and centered response, so no intercept is displayed or penalized. Ridge minimizes

$$J_{\lambda(\beta)} = \|y - X\beta\|^2 + \lambda\|\beta\|^2, \quad \lambda > 0,$$

and has the registered estimator

$$\hat{\beta}_\lambda = (X^T X + \lambda I_d)^{-1} X^T y.$$

The Hessian is $2(X^T X + \lambda I_d)$ and is positive definite, so the coefficient minimizer is unique even for dependent columns.

Uniqueness alone does not certify lower prediction risk.

Scaling: if a source uses $\frac{\|y - X\beta\|^2}{n} + \lambda' \|\beta\|^2$, the same family uses $\lambda = n\lambda'$. Do not mix their numerical parameter values. An unpenalized intercept may equivalently be fitted before centering; penalizing it is a different modeling convention.

2. Rank and selection boundaries

With compact SVD $X = U_r \Sigma_r V_r^T$,

$$\hat{\beta}_\lambda = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \lambda} \langle u_j, y \rangle v_j.$$

Hence $\hat{\beta}_\lambda \rightarrow X^+ y$ as $\lambda \rightarrow 0^+$. Under full column rank, this is $(X^T X)^{-1} X^T y$; under rank deficiency it is the minimum-norm OLS minimizer, not an arbitrary OLS solution. As $\lambda \rightarrow \infty$, every displayed coefficient tends to zero.

Ridge continuously shrinks generic coefficients and does not search feature subsets or normally set coefficients exactly to zero. A coefficient in an exact design-null direction is zero because it cannot improve fit but still pays the penalty—not because ridge performed variable selection.

4. Directional bias and variance

1. Gram eigendirections and shrinkage

Let

$$G = X^T X = Q \operatorname{diag}(\mu_1, \dots, \mu_d) Q^T, \quad \mu_j \geq 0,$$

where μ_j are squared singular values and $z_j = \langle q_j, X^T y \rangle$. Ridge uses coefficient $\frac{z_j}{\mu_j + \lambda}$. For $\mu_j > 0$, relative to OLS it retains

$$s_{j(\lambda)} = \frac{\mu_j}{\mu_j + \lambda}.$$

Small- μ_j directions shrink most. Keep three quantities distinct:

- $\frac{1}{\mu_j + \lambda}$ acts on z_j ;
- $s_{j(\lambda)}$ is the fraction of the OLS coefficient retained;
- if $\mu_j = 0$, ridge assigns zero in that coefficient direction.

2. Fixed-design probability model

To discuss moments, declare

$$y = X\beta^* + \varepsilon, \quad \mathbb{E}[\varepsilon|X] = 0, \quad \operatorname{Cov}(\varepsilon|X) = \sigma^2 I_n,$$

with fixed X and a correctly specified linear conditional mean. Then

$$\mathbb{E}[\hat{\beta}_\lambda|X] = (G + \lambda I)^{-1} G\beta^*,$$

$$\operatorname{Cov}(\hat{\beta}_\lambda|X) = \sigma^2 (G + \lambda I)^{-1} G (G + \lambda I)^{-1}.$$

For $\theta_j = \langle q_j, \beta^* \rangle$,

$$\mathbb{E}[\hat{\theta}_{\lambda,j}|X] = s_{j(\lambda)} \theta_j,$$

$$\text{bias}_j = -\frac{\lambda}{\mu_j + \lambda} \theta_j,$$

$$\text{variance}_j = \sigma^2 \frac{\mu_j}{(\mu_j + \lambda)^2}.$$

3. Coefficients are not predictions

Coefficient and prediction risks differ. For fitted values $X\hat{\beta}_\lambda$, directional squared-bias and variance gain μ_j ; sum and divide by n for in-design mean error, while fresh-response error adds noise. Training residuals

determine none; λ trades bias for sensitivity, so risk need not be monotone.

5. Diagnose before regularizing

1. Evidence-to-intervention map

Equal training error can hide different failures. Combine variation across refits, Gram-spectrum conditioning, systematic prediction bias, fresh-noise scale, and the scientific prediction goal.

Dominant evidence	First response
high sample variance, no weak-direction evidence	more data or another justified variance control; ridge is not automatic
variance aligned with tiny μ_j	ridge is targeted; monitor its added bias
high systematic bias	enrich the representation/model, not stronger shrinkage
high fresh-noise floor	improve measurement or accept the irreducible floor

In every case, later independent evidence should estimate out-of-sample predictive risk and be interpreted against the diagnosed failure/noise floor.

2. Diagnosis, selection, and final testing

Diagnosis says what λ must trade off; it does not choose its value. Fit candidates on training data, select using honest out-of-sample evidence or another justified criterion, then use the untouched final test set once. Tuning on final-test error makes the reported minimum optimistically biased.

The penalty is an SRM-style fit-complexity compromise across $\mathcal{H}_B = \{x \mapsto \beta^T x : \|\beta\| \leq B\}$. It does not literally lower the VC dimension of an unchanged class: a penalty changes the criterion; a class restriction changes the hypothesis set.